INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALTLAKE
(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA
Term – I , Third Semester Examination, August- 2025
Subject Code: BCACC301
Subject Name: Design and Analysis of Algorithms

Full Marks: 30 Duration: 1 Hour

Part - A
Answer any 5 questions out of 6
Each question carries 2 marks (2 × 5)

1. Explain the concept of time complexity and its significance in algorithm analysis. **2**
2. What is the least number of nodes possible in an almost complete binary tree of height *h*? **2**
3. What is the purpose of the `Partition` procedure in Quicksort? **2**
4. Find out the time complexity of the following algorithm **2**

```
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for(i=1;i<n;i=i*2){
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}
```

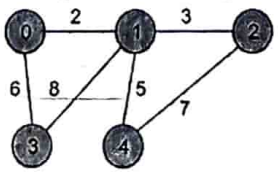
5. Solve using master theorem: T(n)=16T(n/4)+n! **2**
6. What is a greedy algorithm? Give one example of a problem that can be solved using this approach. **2**

Part - B
Answer any 2 questions out of 3
Each question carries 5 marks (5 × 2)

7. Differentiate between Big-O, Big-Theta (Θ), and Big-Omega (Ω) notations with proper mathematical definitions and examples. Why Θ notation is considered the tight bound? **5**
8. Evaluate this recurrence relation T(n) = 2T($\sqrt{n}$)+log n using Master Theorem **5**
9. List and explain the fundamental properties of Asymptotic Notations. **5**

Part - C
Answer any 1 question out of 2
Each question carries 10 marks (10 × 1)

10. Given a connected, undirected graph with V vertices and E weighted edges, compute the total weight of the edges included in its Minimum Spanning Tree (MST). **10**



11. Design a divide and conquer algorithm to find the maximum and minimum elements in an array of n numbers. **10**

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